Heart Disease Prediction Using

Logistic Regression and Clinical Features

The Problem

- Heart disease is a leading cause of mortality worldwide.
- Clinical diagnostic procedures involve complex evaluations and potential risks.
- Accurately predicting heart disease presence using patient clinical attributes remains a critical healthcare challenge.

Why It Matters

- Healthcare providers and cardiologists can use these predictive insights to improve patient clinical outcomes and diagnostic accuracy.
- Identifying key risk factors (such as chest pain type, ST-slope, and thalassemia) enables early targeted interventions and research into underlying cardiovascular causes.

Our Approach

- **Data Preparation:** Clean and organize the heart disease dataset, removing 723 duplicate records to yield 302 unique observations and prevent data leakage.
- **Visualization:** Use graphs and charts to describe dataset distributions and clinical attribute interactions.
- **Exploratory Data Analysis (EDA):** Investigate numerical and categorical attribute relationships, correlation structures, and clinical trends
- **ML Model Development:** Build and evaluate supervised classification models (Linear Probability Model and Logistic Regression) to predict heart disease presence with high diagnostic accuracy.

Data information

- **Data source:** Heart Disease Dataset from the UCI Machine Learning Repository
- **Study area:** Cleveland Clinic Foundation
- **Number of Samples:** 302 unique patient observations (cleaned from 1,025 raw records by removing duplicates)

- **Data Characteristics:**

– Balanced dataset: 164 patients with presence of heart disease (54.3%), 138 patients with absence (45.7%)
– Mixed data types: Numerical, Categorical, Binary

Variables:

Category	Variable Name	Type
Demographics	Age	Numerical
	Sex	Binary
Clinical Measurements	Resting Blood Pressure	Numerical
	Serum Cholesterol	Numerical
	Maximum Heart Rate (Max-heart-rate)	Numerical
	Fasting Blood Sugar	Binary
Diagnostic Tests	Resting ECG	Categorical
	ST Depression (old peak)	Numerical
	Slope of Peak Exercise ST Segment (slope)	Categorical
	Major Vessels by Fluoroscopy	Categorical
	Thalassemia	Categorical
Symptoms & Stress Test	Chest Pain Type	Categorical
	Exercise Induced Angina	Binary
Target Outcome	Heart Disease Presence (target)	Binary

Data Pre-processing:

- **Encoding Variables：**Converted categorical text variables (sex ,chest-pain-type ,thalassemia) into standard R factor levels for proper statistical modeling

- **Missing Values:** No missing values in the dataset

- **Handling Duplicates:** Identified and removed 723 duplicate rows from the initial dataset (reduced from 1,025 to 302 unique observations) to prevent model overfitting.

- Before Duplicate Removal (N=723)

```
Summarize numeric variables:
                      n_obs n_missing n_distinct    mean  median     min     max     p25     p75     sd    se
                    age   723         0         41  54.440  56.000  29.000  77.000  48.000  61.000  9.089 0.338
 resting_blood_pressure   723         0         49 131.615 130.000  94.000 200.000 120.000 140.000 17.509 0.651
            cholestoral   723         0        152 245.791 240.000 126.000 564.000 211.000 275.500 51.560 1.918
         Max_heart_rate   723         0         91 148.924 152.000  71.000 202.000 132.000 166.000 23.061 0.858
                oldpeak   723         0         40   1.083   0.800   0.000   6.200   0.000   1.800  1.181 0.044
                 target   723         0          2   0.501   1.000   0.000   1.000   0.000   1.000  0.500 0.019

Summarize factors:
     sex              chest_pain_type             fasting_blood_sugar                        rest_ecg   exercis
 Female:216   Asymptomatic    : 54    Greater than 120 mg/ml:108      Left ventricular hypertrophy: 11   No :477
 Male  :507   Atypical angina :117    Lower than 120 mg/ml  :615      Normal                       :350   Yes:246
              Non-anginal pain:198                                    ST-T wave abnormality        :362
              Typical angina  :354
```

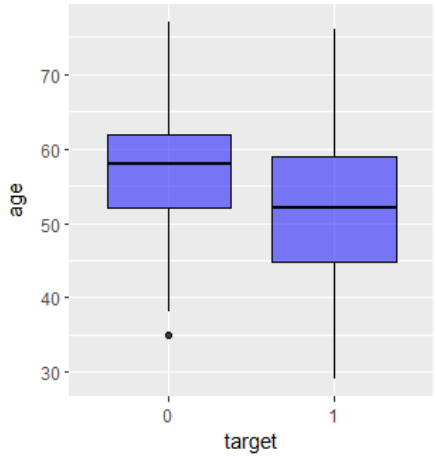
- After Duplicate Removal (N=302)

```
Summarize numeric variables:
                      n_obs n_missing n_distinct    mean  median     min     max     p25     p75     sd    se
                    age   302         0         41  54.421  55.500  29.000  77.000  48.000  61.000  9.048 0.521
 resting_blood_pressure   302         0         49 131.603 130.000  94.000 200.000 120.000 140.000 17.563 1.011
            cholestoral   302         0        152 246.500 240.500 126.000 564.000 211.000 274.750 51.753 2.978
         Max_heart_rate   302         0         91 149.570 152.500  71.000 202.000 133.250 166.000 22.904 1.318
                oldpeak   302         0         40   1.043   0.800   0.000   6.200   0.000   1.600  1.161 0.067
                 target   302         0          2   0.543   1.000   0.000   1.000   0.000   1.000  0.499 0.029

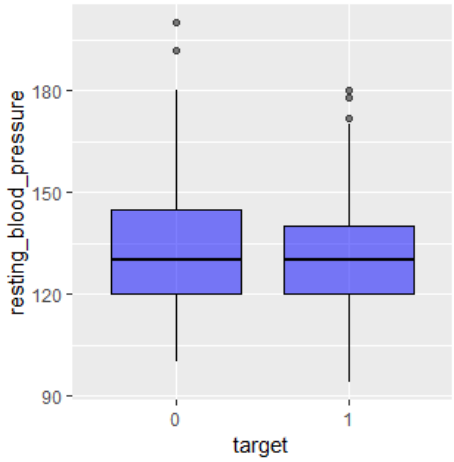
Summarize factors:
     sex              chest_pain_type             fasting_blood_sugar                        rest_ecg   exercis
 Female: 96   Asymptomatic    : 23    Greater than 120 mg/ml: 45      Left ventricular hypertrophy:  4   No :203
 Male  :206   Atypical angina : 50    Lower than 120 mg/ml  :257      Normal                       :147   Yes: 99
              Non-anginal pain: 86                                    ST-T wave abnormality        :151
              Typical angina  :143
```

- **Outliers & Distribution Analysis:**

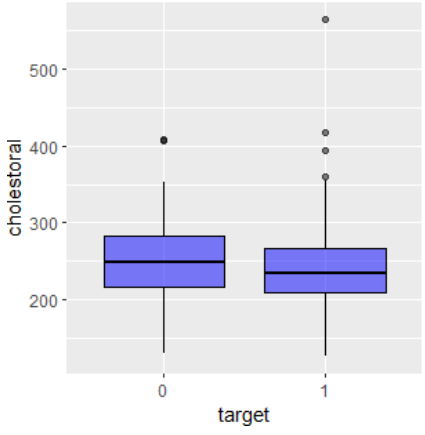
Age Distribution: Heart disease patients show a slightly lower median age



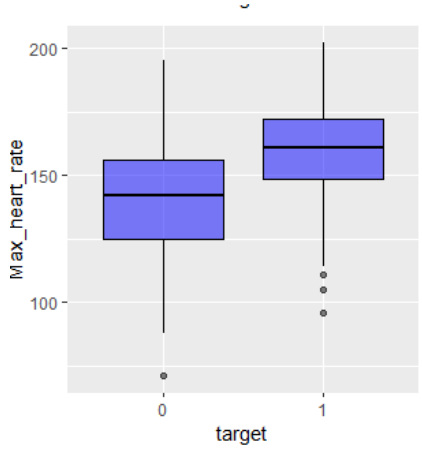
Resting Blood Pressure: Identifies minor upper outliers (170–200 mmHg).



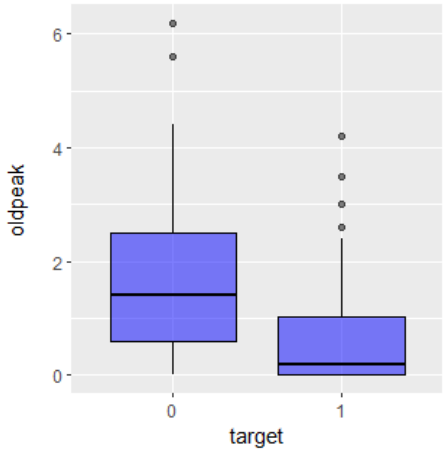
Cholesterol: Shows a severe hypercholesterolemia outlier (~564 mg/dl)



Max Heart Rate: Higher median heart rate observed in group 1

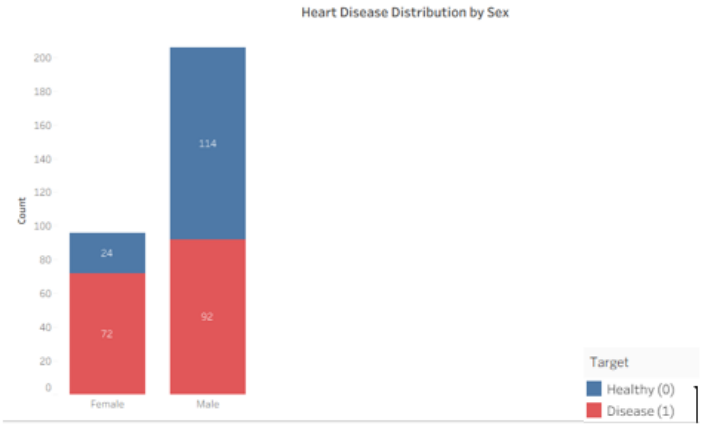


ST Depression (Old peak): Healthy group exhibits higher overall Depression



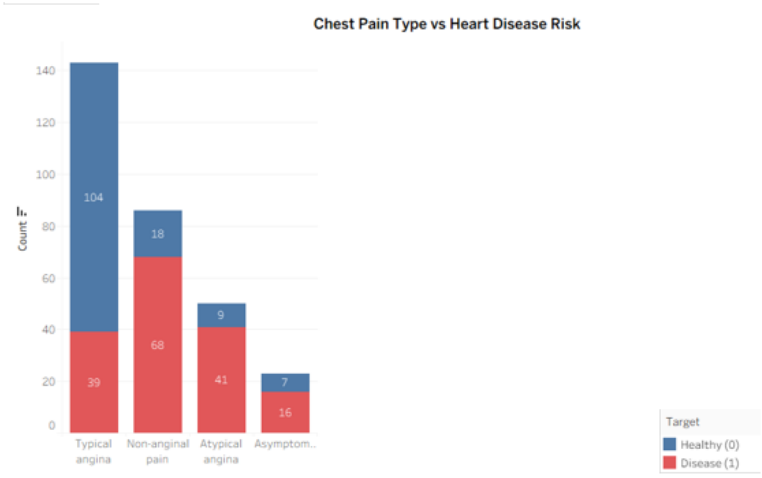
Boxplot analysis indicated several outliers in continuous variables such as resting blood pressure and cholesterol. Since these extreme measurements are clinically valid and represent true high-risk patient profiles, they were intentionally retained. Preserving these data points prevents the loss of critical information and ensures the model accurately reflects real-world clinical scenarios."

Data Visualization:



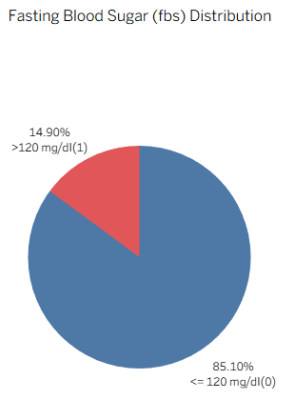
Heart Disease Distribution by Sex

This chart illustrates the distribution of heart disease across genders. As observed, the proportion of affected male cases is higher in the sample data, which justifies the inclusion of sex as a fundamental demographic factor in the predictive modeling.



Chest Pain Type vs Heart Disease Risk

This stacked bar chart displays the distribution of different chest pain types among healthy and diseased individuals. The significant concentration of cases within specific chest pain categories highlights the critical role of this variable as a primary clinical indicator of ischemia.



Fasting Blood Sugar Distribution

This pie chart shows the relative frequency of fasting blood sugar levels (FBS), where approximately 85% of the data points fall at or below 120 mg/dl. This visualization clearly demonstrates why FBS was ultimately excluded during stepwise screening due to a lack of statistical significance in the final model

Overview of Analysis

Objective:

To predict **Heart Disease Status (target)** (1 = Disease Present, 0 = No Disease) using various statistical and predictive modeling techniques based on patient clinical attribute

Models & Methods:

Correlation Analysis

Linear Regression (OLS Baseline)

Logistic Regression

Correlation Analysis

Interpretation:

Correlation plot describe how these variables relate to each other.

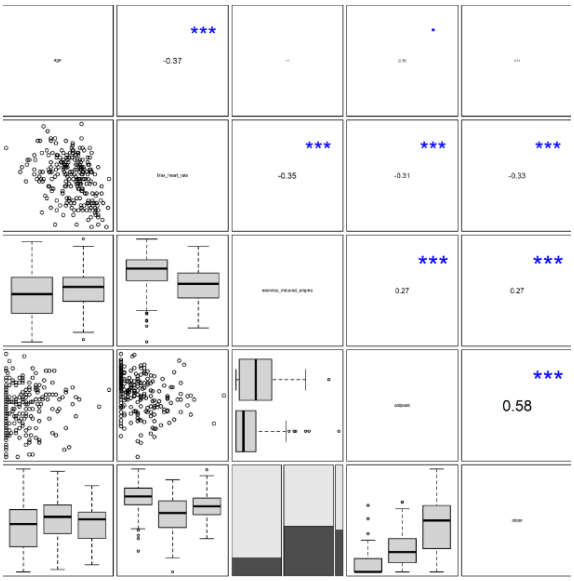
Full Dataset Screening:

Initial correlation matrix evaluation across all 13 dataset attributes confirmed that no variables have a high correlation with any other less than 0.7

All pairwise correlation values fall well below the critical threshold (0.8 /0.7) confirming the absence of severe multicollinearity across the entire dataset.

Focused Analysis on Key Clinical Relationships:

To provide a clearer view, a focused correlation plot was generated for the subset of variables exhibiting the strongest relative correlations (slope, old peak, exercise-induced-angina, Max-heart-rate, and Age):



Strongest Positive Correlation — Slope vs. old peak (r = 0.58):

This positive trend makes intuitive clinical sense, as both parameters directly reflect exercise-induced ST-segment alterations during ECG stress testing.

Moderate negative correlation between Exercise-Induced Angina and Max Heart Rate (r = -0.35):

indicating that patients who achieve higher heart rates are less likely to experience exercise angina**.**

Mild negative trend for age against maximum heart rate (r = -0.37):

makes intuitive sense as it would be expected that as you get older, your maximum achievable heart rate decreases.

Linear Regression

Not Suitable for Categorical Data

Model 1 – Linear Regression (OLS)

Dependent Variable: target (Heart Disease)
Results: R^2 = 0.575, Adjusted R^2 = 0.532

```
                                      coefficient std.error t.value p.value
(Intercept)                                 0.887     0.473   1.875   0.062 .
age                                         0.003     0.003   1.042   0.299
sex|Male                                   -0.137     0.055  -2.470   0.014 *
chest_pain_type|Atypical angina            -0.078     0.103  -0.762   0.447
chest_pain_type|Non-anginal pain           -0.009     0.094  -0.097   0.923
chest_pain_type|Typical angina             -0.231     0.092  -2.504   0.013 *
resting_blood_pressure                     -0.001     0.001  -1.014   0.312
cholestoral                                -0.000     0.000  -0.056   0.956
fasting_blood_sugar|Lower than 120 mg/ml   -0.072     0.069  -1.056   0.292
rest_ecg|Normal                            -0.041     0.208  -0.199   0.843
rest_ecg|ST-T wave abnormality              0.007     0.209   0.035   0.972
Max_heart_rate                              0.001     0.001   1.098   0.274
exercise_induced_angina|Yes                -0.086     0.057  -1.517   0.131
oldpeak                                    -0.054     0.026  -2.040   0.043 *
slope|Flat                                 -0.167     0.055  -3.021   0.003 **
slope|Upsloping                            -0.076     0.111  -0.683   0.495
vessels_colored_by_flourosopy|One          -0.165     0.253  -0.653   0.515
vessels_colored_by_flourosopy|Three        -0.219     0.265  -0.827   0.409
vessels_colored_by_flourosopy|Two          -0.179     0.262  -0.685   0.494
vessels_colored_by_flourosopy|Zero          0.146     0.250   0.583   0.561
thalassemia|No                             -0.306     0.248  -1.233   0.219
thalassemia|Normal                         -0.072     0.103  -0.700   0.485
thalassemia|Reversable Defect              -0.201     0.057  -3.515  < .001 ***

Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

R-squared: 0.575,  Adjusted R-squared: 0.532
F-statistic: 13.408 df(22,218), p.value < .001
Nr obs: 241
```

This model is not appropriate for categorical outcome leading to meaningless interpretation

Feature Selection:

To mitigate the risk of overfitting caused by irrelevant or noisy variables, the feature selection process was conducted using a hybrid approach that combined clinical domain knowledge with a stepwise selection algorithm. Initially, guided by cardiovascular literature and established diagnostic standards, critical clinical features associated with myocardial ischemia and cardiac performance were pre-screened. Subsequently, employing a stepwise selection procedure based on the Akaike Information Criterion (AIC), the model systematically eliminated variables that failed to contribute significantly to predictive performance (such as age, serum cholesterol, and resting blood pressure), thereby avoiding unnecessary model complexity.

Included Variables:

- **Chest Pain Type:**
  The strongest initial clinical predictor in cardiac triage, which the Stepwise algorithm retains in the early stages.
- **Maximum Heart Rate:**
  Represents the heart's physiological response to stress or exercise; a lower maximum heart rate is associated with a higher risk of heart disease
- **Exercise-Induced Angina:**
  A direct indicator of insufficient oxygen supply to the myocardium during physical activity
- **Old Peak (ST-Segment Depression) and slope (ST-Segment Slope):**
  Key electrocardiographic indicators obtained from an exercise stress test that provide important information for assessing cardiovascular abnormalities
- **Number of Major Vessels:**
  An important imaging finding obtained through fluoroscopy/angiography that provides information about the condition of the major coronary vessels
- **Thalassemia:**
  A diagnostic variable in this dataset that provides information related to blood flow and cardiac function.
- **Sex:**
  A basic demographic factor associated with differences in cardiovascular disease risk between males and females

Excluded Variables:

- **Age and Resting Blood Pressure:**
  Clinically, these are considered long-term risk factors. However, in this relatively small sample, when more direct diagnostic variables, such as exercise-test responses and the number of affected coronary vessels, are included in the model, the effects of age and resting blood pressure may be diminished. The Stepwise algorithm excludes these variables because they do not provide a significant improvement in the AIC, thereby helping to reduce unnecessary model complexity and the risk of overfitting.

- **Cholesterol and Fasting Blood Sugar:**
  These are general biochemical measurements that have an indirect association with heart disease. In the presence of more direct diagnostic variables, such as angiography (vessels-colored), their predictive contribution in the multivariable model decreases. As a result, the stepwise selection procedure identifies them as variables that can be excluded while optimizing the model based on AIC.

- **Resting ECG:**
  In this dataset, the resting ECG results generally do not show substantial abnormalities or strong variations among patients. Therefore, its discriminatory power is lower than that of exercise-related variables such as (Old Peak and Max-heart-rate). For this reason, Rest-ECG is removed during the first step of the Stepwise variable selection process.

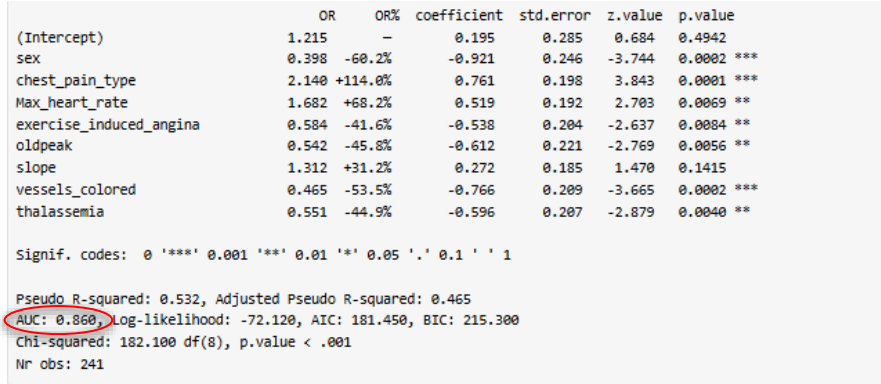
Logistic Regression:

- **Train Model**

Objective:
Logistic regression applied to train the model to predict target (Heart Disease) in Training Dataset and test the model in Testing Dataset.

Data Sets:
Training Data (241)
Test Data (61)

Key Metrics:
Accuracy
AUC
Model coefficients



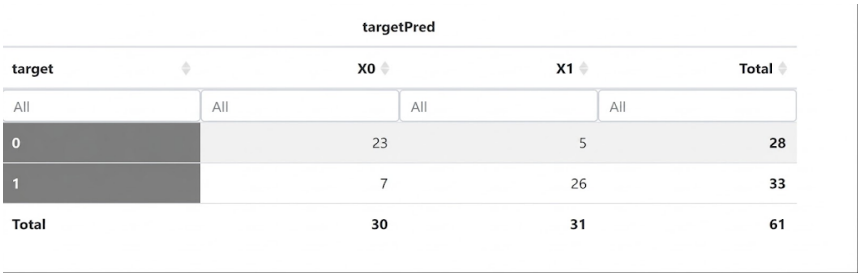
	OR	OR%	coefficient	std.error	z.value	p.value	
(Intercept)	1.215	–	0.195	0.285	0.684	0.4942	
sex	0.398	-60.2%	-0.921	0.246	-3.744	0.0002	***
chest_pain_type	2.140	+114.0%	0.761	0.198	3.843	0.0001	***
Max_heart_rate	1.682	+68.2%	0.519	0.192	2.703	0.0069	**
exercise_induced_angina	0.584	-41.6%	-0.538	0.204	-2.637	0.0084	**
oldpeak	0.542	-45.8%	-0.612	0.221	-2.769	0.0056	**
slope	1.312	+31.2%	0.272	0.185	1.470	0.1415	
vessels_colored	0.465	-53.5%	-0.766	0.209	-3.665	0.0002	***
thalassemia	0.551	-44.9%	-0.596	0.207	-2.879	0.0040	**

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Pseudo R-squared: 0.532, Adjusted Pseudo R-squared: 0.465
AUC: 0.860, Log-likelihood: -72.120, AIC: 181.450, BIC: 215.300
Chi-squared: 182.100 df(8), p.value < .001
Nr obs: 241

Confusion Matrix Evaluation

- **Test Model**



	targetPred		
target	X0	X1	Total
All	All	All	All
0	23	5	28
1	7	26	33
Total	30	31	61

(Threshold =0.5)

True Positives (TP = 26): Correctly identified 26 patients with heart disease**.**

True Negatives (TN = 23): Correctly identified 23 healthy individuals**.**

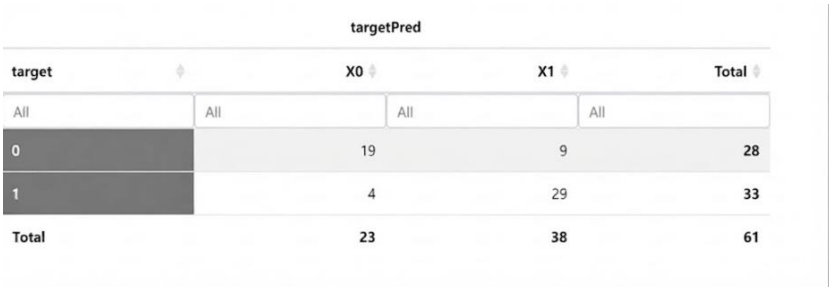
False Positives (FP = 5): Misclassified 5 healthy individuals as having heart disease

False Negatives (FN = 7): Misclassified 7 heart disease patients as healthy

Accuracy = (23+26)/61 = 0.803 ⟶ 80.3%

Specificity = 23/ (23+5) = 0.821 ⟶ 82.1%

Sensitivity = 26/ (7+26) = 0.788 ⟶ 78.8%



	targetPred		
target	X0	X1	Total
All	All	All	All
0	19	9	28
1	4	29	33
Total	23	38	61

(Threshold =0.3)

Accuracy = (19+29)/61 = 0.787 ⟶ 78.7%

Specificity = 19/ (19+9) = 0.679 ⟶ 67.9%

Sensitivity = 29/ (4+29) = 0.879 ⟶ 87.9%

Analysis: Lowering the threshold to 0.3 increases Sensitivity (detecting more heart disease patients), but reduces Specificity

Logistic Regression - Insights from Results

- **High Accuracy (80.3%)**: Influenced by a reasonably balanced distribution between positive and negative cases in the dataset.

- **Balanced Specificity (82.1%):** Model is reliable at predicting individuals who will not face the risk of heart disease.

- **Moderate Sensitivity (78.8%)**: The model successfully captures a large portion of patients at risk, though a few cases are still missed.

Strengths:

Provides a solid overall performance and is beneficial if the focus is on achieving a stable balance between identifying healthy individuals and detecting actual cases.

Weakness / Trade-off:

Failing to identify some patients at risk (7 False Negatives in the test set) highlights a critical limitation in a healthcare context, where missing a sick patient carries a high cost.

This indicates that the default threshold (0.5) might require tuning (lowering to 0.3) to prioritize sensitivity and ensure fewer high-risk patients are misclassified as healthy.

Strategies for Improved Results

- Explore advanced hyperparameter tuning for the logistic regression model or benchmark against tree-based ensembles (e.g., Random Forest, XG Boost) to evaluate whether non-linear algorithms capture complex interactions among the retained clinical features.

- Investigate feature interaction terms—particularly between electro-diagnostic indicators like Old Peak and slope—to model non-linear physiological responses more explicitly.

- Conduct clinical cost-benefit analysis by adjusting the decision classification threshold to minimize false negatives, prioritizing patient safety in cardiac disease screening.